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Electric and magnetic properties of LiH $X\,^1\Sigma^+$ from analysis of pure rotational and vibration–rotational spectra

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Abstract

Pure rotational and vibration–rotational transitions (1530 lines) of LiH $X\,^1\Sigma^+$ in four isotopic variants are analyzed with a deformationally self-consistent procedure in an analytic version. Using a relation between nonadiabatic rotational effects and electric and magnetic properties of the molecule, we estimate a permanent dipole moment of $\mu_0 = 1.933(76) \times 10^{-29}$ Cm, electric polarity $^+\mathrm{LiH}^-$ and rotational g-factor $g_0 = -0.253(14)$ for $^7\mathrm{Li}\,^2\mathrm{H}$ and $g_0 = -0.618(28)$ for $^7\mathrm{Li}\,^1\mathrm{H}$. They conform acceptably with magnitudes from published experiments on Stark and Zeeman effects and from quantum-chemical calculations of electronic structure. © 1999 Elsevier Science B.V. All rights reserved.

A vibrating and rotating polar diatomic molecule is endowed with electric and magnetic dipole moments

$$\boldsymbol{\mu} = \sum_{j=1}^{N+2} q_j \boldsymbol{r}_j\,, \quad \boldsymbol{m}_J = \mu_n g_J \boldsymbol{J}/\hbar\,, \tag{1}$$

associated with electric and magnetic fields generated by moving constituents of the molecule, i.e. N electrons and two nuclei. The electric moment depends on the spatial distribution $\boldsymbol{r}_j$ of charges q_j in the molecule, whereas the magnetic moment is proportional to the angular momentum $\boldsymbol{J}$ of the system; in the absence of electronic orbital and spin and of nuclear contributions to the total angular momentum, the proportionality factor includes the nuclear magneton $\mu_n = e\hbar/2m_\mathrm{p}$ (e is protonic charge, m_p protonic mass), and a dimensionless but signed quantity g_J called the rotational g-factor. $\boldsymbol{\mu}$ depends upon coordinates of both electrons and nuclei, but commonly in use is a function $\mu(R)$ for electric dipole moment in a particular electronic state, which depends on an instantaneous internuclear separation R. It is expressed as a power series [1]

$$\mu(R) = \sum_{i=0} \mu_i x^i\,, \quad x = (R - R_0)/R_0\,, \tag{2}$$

in terms of a reduced internuclear displacement from the equilibrium separation [1] R_0, called Dunham's variable [2]. The coefficient μ_0 in (2) is a permanent electric moment and μ_1 is the first term involved in a vibrationally modified dipole moment of the system. The former governs the intensity of the pure rotational band in absorption or emission, whereas the latter analogously influences the intensity of the fundamental band $v = 0 \rightarrow 1$.

In a neutral linear molecule, the direction of the permanent dipole moment coincides with the inter-

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[1] In this work, we use the notation R_0, μ_0 instead of the standard R_e, μ_e referring to quantities in the state with $J = 0$ and $v = 0$.

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nuclear axis and its magnitude is independent of the origin of a coordinate system. For ions the magnitude of the dipole moment depends on the location of the origin of a coordinate system, generally placed at the centre of mass of a molecule.

The rotational g-factor, in a particular electronic state, varies only slightly with total overall angular momentum $\boldsymbol{J}$, but it depends on the nuclear masses m_a, m_b and the internuclear separation R. Hence one introduces a representation for $g_J(R)$ as a power series in an analogous form to (2)

$$g_J(R) = \sum_{i=0} g_i x^i. \tag{3}$$

The primary sources of information about μ and g_J are experiments on the Stark and Zeeman effects. In the microwave range, the former yields only a magnitude of the permanent dipole moment; although the latter can provide not only the magnitude but also the sign of the rotational g-factor g_J [3], the sign is generally assumed. Then the sign of μ_0 is estimated from a relation valid for neutral molecules

$$\mu_0 = \frac{-e(I'g'_J - Ig_J)}{2m_p\Delta R}, \tag{4}$$

in which I' and g'_J denote the moment of inertia I and rotational g-factor of an isotopically substituted species; such a substitution alters the centre of molecular mass by an increment ΔR.

The most accurate results are obtained in Stark and Zeeman experiments combining a molecular beam with electric [4,5] or magnetic [6] resonance; in the latter case the method might fail to yield a reliable value of g_J [7].

For LiH in the electronic ground state $X\,^1\Sigma^+$, highly resolved molecular-beam electric-resonance spectra were observed by Rothstein [5], for one rotational and two vibrational states; he estimated [2] permanent dipole moments are $\mu_0 = 1.94393(21) \times 10^{-29}$ Cm for $^7Li\,^1H$, $\mu_0 = 1.94433(66) \times 10^{-29}$ Cm for $^6Li\,^1H$, and $\mu_0 = 1.94399(53) \times 10^{-29}$ Cm for $^6Li\,^2H$.

Application of the molecular-beam magnetic-resonance method to LiH $X\,^1\Sigma^+$ by Lawrence et al. [6] yielded a rotational g-factor $g_J = -0.654(7)$ for $^7Li\,^1H$, $g_J = -0.272(5)$ for $^7Li\,^2H$, and a rough value of the permanent dipole moment $\mu_0 = 1.97(17) \times 10^{-29}$ Cm at assumed polarity $^+LiH^-$. More accurate measurements were made by Freeman et al. [4]; from molecular Zeeman spectra of LiH $X\,^1\Sigma^+$, obtained in the first rotational and zero vibrational states by molecular beam electric resonance, they determined $g_J = -0.65842(17)$ for $^7Li\,^1H$ and $g_J = -0.27674(11)$ for $^7Li\,^2H$.

A secondary source of dipole moment and rotational g-factor is quantum-chemical calculation, which provides not only a value for a particular vibration–rotational state, but also dependence on R [8–10]. The electronic computation of μ and g_J has the disadvantage of producing a result that depends strongly on the method and its level of approximation [9,10]; hence which calculated value best represents the pertinent quantity is difficult to assess. However, in the absence of an experimental value, a quantum-chemical calculation is necessarily the only source of information on μ and g_J.

Application of the method of polarization propagation to $^7Li\,^1H$ $X\,^1\Sigma^+$, in the first rotational and zero vibrational states, yielded $g_0 \in \langle -0.6898, -0.6351\rangle$ for various levels of approximation [10]. In particular, the modified coupled-cluster singles and doubles polarization propagator approximation (CCSDPPA) [10] provided expansions in explicit form (2) and (3) [12] in terms of Ogilvie's variable $z = 2(R - R_0)/(R + R_0)$ [11].

A third source of μ and g_J is the direct inversion of highly resolved pure rotational and vibration–rotational spectra [8,12,13], particularly those obtained with Fourier-transform spectrometers and diode lasers.

For the purpose of such an analysis one solves Schrödinger's equation [12,14]

$$\left(-\frac{\hbar^2}{2m}\frac{d^2}{dR^2} + U_{vJ}(R) - E_{vJ}\right)\psi_{vJ}(R) = 0, \tag{5}$$

$$U_{vJ}(R) = \frac{\hbar^2 J(J+1)[1 + \alpha(R) - \beta(R)]}{2mR^2} + V(R)[1 - \beta(R)] + V'(R) + E_{vJ}^{BO}\beta(R), \tag{6}$$

with adiabatic $V'(R)$ and nonadiabtic rotational $\alpha(R)$ and vibrational $\beta(R)$ corrections to take account of the fact that electrons fail to follow perfectly the

[2] Values of μ_0 are converted to SI units using the factor 1 debye = 3.33564×10^{-30} Cm; errors propagate from experimental uncertainties of dipole moments quoted by Rothstein.

nuclei in their vibrational and rotational motions; thus energies deviate from E_{vJ}^{BO}.

All radial functions in (5) are expanded into a series in Ogilvie's variable,

$$V(R) = c_0 z^2\left(1 + \sum_{i=1} c_i z^i\right),$$

$$V'(R) = m_e \sum_{i=1} \sum_{\mu=a,b} u_i^{\mu} m_{\mu}^{-1} z^i, \tag{7}$$

$$\alpha(R) = m_e \sum_{i=0} \sum_{\mu=a,b} t_i^{\mu} m_{\mu}^{-1} z^i,$$

$$\beta(R) = m_e \sum_{i=0} \sum_{\mu=a,b} s_i^{\mu} m_{\mu}^{-1} z^i. \tag{8}$$

In Eqs. (5) and (6), m is the reduced mass of the system; m_e is the electronic mass.

To solve the wave equation (5), one expands (7) and (8) into a series of the Dunham's variable $x = (R - R_0)/R_0$, and then applies the hypervirial perturbative approach [15] producing eigenenergies [12]

$$E_{vJ} = \sum_{k,l=0} \sum_{\mu=a,b} \left(Y_{kl} + Z_{kl}^{r,\mu} + Z_{kl}^{v,\mu}\right) \times (v + 1/2)^k \left[J(J+1)\right]^l, \tag{9}$$

in which Y_{kl} are Dunham's term coefficients depending on R_0 and c_i, whereas $Z_{kl}^{r,a(b)}$ and $Z_{kl}^{v,a(b)}$ represent adiabatic and rotational and vibrational nonadiabatic terms depending on $u_i^{a,b}$, $t_i^{a,b}$ and $s_i^{a,b}$.

Coefficients of radial functions (7) and (8) are directly fitted to measured rotational and vibration–rotational spectra [12] by means of the eigenenergies (9). The electric dipole moment and rotational g-factor can be evaluated from the relationships [10]

$$\mu(R) = (eR/2) \sum_{i=0} \left(t_i^{\mathrm{a}} - t_i^{\mathrm{b}}\right) z^i,$$

$$g_J(R) = m_p \sum_{i=0} \sum_{\mu=a,b} t_i^{\mu} m_{\mu}^{-1} z^i, \tag{10}$$

including parameters $t_i^{a,b}$ to represent nonadiabatic rotational effects.

This method applied to heavy molecules such as AlH, GeS, GaH, or BrCl in the $X\,^1\Sigma^+$ or 0^+ ground electronic state provided g_J [8,12,13] and μ_0 [12,13], which conforms acceptably with their values obtained from experiment or quantum-chemical computations. For the light molecule LiH, the estimation of μ_0 and g_J directly from the spectra was impracticable because strong correlation between the radial parameters in the sets (t_0^{Li}, s_0^{Li}, u_1^{Li}) and (t_0^{H}, s_0^{H}, u_1^{H}) precluded the satisfactory evaluation of $t_0^{\mathrm{Li,H}}$. Hence the permanent electric dipole moment and rotational g-factor defined according to [12]

$$\mu_0 = eR_0\left(t_0^{\mathrm{a}} - t_0^{\mathrm{b}}\right)/2, \quad g_0 = m_p\left(t_0^{\mathrm{a}}/m_a + t_0^{\mathrm{b}}/m_b\right), \tag{11}$$

become indeterminate.

In the present work, μ_0 and g_0 for LiH $X\,^1\Sigma^+$ are evaluated from pure rotational and vibration–rotational spectra using an alternative approach to those applied by Maki et al. [16], by Tiemann and Ogilvie [17], by Coxon [18], by Ogilvie [19], by Uehara [20,21] and by Bernath and co-workers [22].

In the calculation, we employ a simplified Schrödinger equation (5)

$$\left[-\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}R^2} + U_J(R) - E_{vJ}(1-\beta_0)\right]\psi_{vJ}(R) = 0, \tag{12}$$

$$U_J(R) = \frac{\hbar^2 J(J+1)\left[1 + \alpha(R) - \beta_0\right]}{2mR^2} + V(R)\left[1 - \beta_0\right] + V'(R), \tag{13}$$

$$\beta_0 = m_e\left(s_0^{\mathrm{a}}/m_a + s_0^{\mathrm{b}}/m_b\right), \tag{14}$$

as no parameters of type $s_{i>0}^{\mathrm{Li,H}}$, pertaining to nonadiabatic vibrational effects of higher order, are evaluable from freely fitted spectral data of LiH at present. E_{vJ}^{BO} in (6) is replaced by E_{vJ}, to restore an original (not approximate) form of this equation [23].

Proceeding along lines of our previous work [24–27], we expand an effective potential (13) in a variable

$$\eta = (R - R_J)/R_J, \tag{15}$$

and then transform (12) to a form

$$\left\{-B_J\frac{\mathrm{d}^2}{\mathrm{d}\eta^2} + a_0^J\eta^2\left(1 + \sum_{s=1} a_s^J\eta^s\right) - \left[E_{vJ}(1-\beta_0) - c_0^J\right]\right\}\psi_{vJ}(\eta) = 0, \tag{16}$$

$$B_J = \hbar^2/\left[2m(R_J)^2\right]$$

$$a_0^J = (2!)^{-1}\left[\mathrm{d}^2U_J(\eta)/\mathrm{d}\eta^2\right]_{\eta=0}, \tag{17}$$

$$a_{s>0}^J = (s!)^{-1}\left[\mathrm{d}^sU_J(\eta)/\mathrm{d}\eta^s\right]_{\eta=0}/a_0^J. \tag{18}$$

$$c_0^J = \left\{B_J J(J+1)\left[1 + \alpha(\eta) - \beta_0\right] + V(\eta)(1-\beta_0) + V'(\eta)\right\}_{\eta=0}, \tag{19}$$

The dynamic reference conformation R_J in (15) is defined according to a criterion for a minimum

$$a_1^J = \left[\frac{\mathrm{d}U_J(\eta)}{\mathrm{d}\eta} \right]_{\eta=0} = 0, \tag{20}$$

which enables the elimination of the linear term $a_1^J\eta$ from (16).

Eq. (16) is amenable to direct solution in a semiclassical (Brillouin–Kramers–Wentzel) scheme to produce eigenvalues in the form

$$E_{vJ} = \left[c_0^J + \sum_{k=0} Y_{k0}(v+1/2)^k \right] / (1-\beta_0); \tag{21}$$

here Y_{k0} are purely vibrational term coefficients [2] into which we substitute $\{R_0, a_n\} \to \{R_J, a_n^J\}$.

In previous work [24–27], we calculated roots R_J of Eq. (20) numerically, with a bisection algorithm, enabling the fit of spectral data in an iterative DS-c scheme. In this work, we apply an analytic version of DS-cP in which R_J are derived by expanding the first derivative of the effective potential (13) into a Taylor series in variable x,

$$f(x) = -\frac{\mathrm{d}U_J(x)}{\mathrm{d}x} = \sum_{k=0} f_k x^k. \tag{22}$$

and then reverting this series to obtain

$$x(f) = \sum_{k=0} f_k^{-1} f^k. \tag{23}$$

According to (20) for $R = R_J$, or equivalently $f(x) = 0$, we derive

$$(R_J - R_0)/R_0 = f_0, \tag{24}$$

so R_J becomes expressed in exact analytic form

$$R_J = R_0(1+f_0). \tag{25}$$

All these operations and calculations [35] of Y_{k0} in (21) and derivatives a_s^J of the effective potential (13) were done using a Maple processor.

The eigenenergies (26) are applied to evaluate parameters of the effective potential (13), and then values of electric dipole moment μ_0 and rotational g-factor g_0 for LiH $X\,^1\Sigma^+$. We expand all radial functions in (13) into Ogilvie's variable which remains finite throughout the range of molecular existence: for $0 \leqslant R < \infty$ we have $-2 \leqslant 2(R-R_0)/(R+R_0) < 2$; this property ensures satisfactory convergence of applied expansions.

To ensure maximal significance of all fitted parameters, we constrained one potential-energy parameter in (7),

$$c_8 = \frac{\mathscr{D}_e}{2^{10}c_0} - \left(2^{-8} + \sum_{i=1}^{7} 2^{-8+i} c_i \right),$$

$$\mathscr{D}_e = 20286\ \mathrm{cm}^{-1} \tag{26}$$

to conform to a relation [28]

$$\mathscr{D}_e = \lim V(R)_{R\to\infty} = 4c_0\left(1 + \sum_{i=1} 2^i c_i\right), \tag{27}$$

and the known dissociation energy $\mathscr{D}_e = 20286\ \mathrm{cm}^{-1}$ of $^7\mathrm{Li}\,^1\mathrm{H}$ [18].

As input data we used 1530 pure rotational and vibration–rotational transitions of $^7\mathrm{Li}\,^1\mathrm{H}$, $^6\mathrm{Li}\,^1\mathrm{H}$, $^7\mathrm{Li}\,^2\mathrm{H}$ and $^6\mathrm{Li}\,^2\mathrm{H}$ including: four [29], six [30], 22 [31,32], 29 [34] and 261 [22] pure rotational lines with $\Delta v = 0$ up to $v = 3$ and $J = 45$; four [33], 526 [16] and 678 [22] rotation–vibrational lines with $\Delta v = 1,2$ up to $v = 6$ and $J = 28$. These spectral data were fitted using a weighted non-linear least-squares routine with weights taken as the inverse squares of the uncertainties of the experimental data. To obtain the best set of parameters fitted from the spectra, we used as criteria the minimum number of fitted parameters consistent with a minimum value of normalized standard deviation $\hat{\sigma} \approx 1$, a maximum value of F-statistic, and optimal values of estimated standard error σ_i of each fitted parameter i and of correlation coefficient $cc(i,j)$ between parameters i and j. The results of the calculations are presented in Table 1. The uncertainty in parentheses is one estimated standard deviation in units of the last quoted digit of values of fitted parameters.

From relations (11) and parameters $t_0^{\mathrm{Li,H}}$ that reflect purely nonadiabatic rotational effects, we evaluate the permanent electric dipole moment and rotational g-factor of two isotopic variants of LiH, reported in Table 2.

The estimated value of $\mu_0 = 1.933(76) \times 10^{-29}$ Cm conforms acceptably (within stated errors) with an experimental value $\mu_0 = 1.94393(21) \times 10^{-29}$ Cm [5] and computed $\mu_0 = 1.9534006(13) \times 10^{-29}$

Table 1
Mass-independent radial parameters of LiH $X\,^1\Sigma^+$

i	c_i	t_i^{H}	$u_i^{\mathrm{Li}}/10^4$ cm^{-1}	$u_i^{\mathrm{H}}/10^4$ cm^{-1}
0	65723.935(84)	−0.731(28)		
1	−0.8971882(73)	2.051(38)	−5.95(32)	−5.43(18)
2	0.348440(27)	−1.697(40)	13.08(75)	
3	−0.09157(14)	1.673(60)	−11.12(73)	4.17(23)
4	−0.04408(71)			
5	0.03486(93)			
6	−0.0197(53)			
7	−0.1196(29)			
t_0^{Li}	0.782(48)			
s_0^{H}	0.590(66)			
$R_0/10^{10}$ m^{-1}	1.59491061(13)			
$\hat{\sigma}$	1.15			
$F/10^{15}$	1.01			

Cm [10]; the rotational g-factor $g_0 = -0.253(14)$ for ^{7}Li ^{2}H and $g_0 = -0.618(28)$ for ^{7}Li ^{1}H are near experimental values $g_J = -0.27674(11)$ and $g_J = -0.65842(17)$ [4], respectively, and conform acceptably with $g_J = -0.2653$ and $g_J = -0.6351$ calculated with the CCSDPPA method [10]. The electric polarity is $^+$LiH$^-$.

Despite strong correlation between parameters quoted in Table 3, the analytic version of DS-cP produces coefficients of radial functions that constitute the most physically meaningful representation of LiH data so far obtained. The present approach enables one to evaluate from infrared and microwave spectra acceptable values of permanent electric dipole moment and rotational g-factor of isotopic variants of LiH $X\,^1\Sigma^+$. Attempts to evaluate these quantities from spectra with alternative methods of reduction seem to have been unsuccessful [16–22].

In particular, Bernath and co-workers [22] analysed similar data including 1476 lines to yield adiabatic and nonadiabatic rotational effects of higher order. They failed to evaluate parameters $t_0^{\mathrm{Li,H}}$, which were constrained to zero during the fit. Such a procedure assumes the lack of a permanent electric dipole moment and rotational g-factor of LiH, which stands in distinct contradiction to experimental evidence. Similar problems were encountered by Uehara [20] who poorly evaluated $t_0^{\mathrm{H}} = -1.09(79)$ with constrained $t_0^{\mathrm{Li}} = 0$; these values incorrectly reproduce μ_0 and g_0. In a subsequent analysis [21], the parameters $t_0^{\mathrm{Li,H}}$ were constrained to zero.

To overcome this difficulty, the spectral data of LiH are generally fitted using $t_0^{\mathrm{Li,H}}$ [17,19] or all $t_i^{\mathrm{Li,H}}$ [10,12] constrained to values consistent either with their relations (11) to known experimental values of electric dipole moment and rotational g-factor, or relations (10) including electronically computed functions (2) and (3). These constraints ensure significant evaluation of fitted parameters, but make the estimation of electric dipole moment and rotational g-factor from the spectra impossible.

Herman and Ogilvie [14] presented a further relation linking nonadiabatic vibrational effects with

Table 2
Electric and magnetic properties of ^{7}Li ^{1}H and ^{7}Li ^{2}H $X\,^1\Sigma^+$

	CAL	EXP[a]	EC[b]
$\mu_0/10^{-29}$ Cm	1.933(76)	1.94393(21)	1.9534006(13)
$g_0^{^7\mathrm{Li}\,^1\mathrm{H}}$	−0.618(28)	−0.65842(17)	−0.6351
$g_0^{^7\mathrm{Li}\,^2\mathrm{H}}$	−0.253(14)	−0.27674(11)	−0.2653

[a] Experimental values from Refs. [4,5].
[b] Electronically calculated values from Ref. [10].

Table 3
Correlation coefficients for strongly correlated parameters

$cc(s_0^{\mathrm{H}},u_1^{\mathrm{H}})$	−1.0000	$cc(t_0^{\mathrm{H}},u_1^{\mathrm{H}})$	−1.0000
$cc(s_0^{\mathrm{H}},u_3^{\mathrm{H}})$	0.9976	$cc(t_0^{\mathrm{H}},u_3^{\mathrm{H}})$	0.9981
$cc(t_0^{\mathrm{Li}},u_1^{\mathrm{Li}})$	−0.9993	$cc(t_0^{\mathrm{H}},s_0^{\mathrm{H}})$	0.9999
$cc(t_0^{\mathrm{Li}},u_2^{\mathrm{Li}})$	0.9932	$cc(u_1^{\mathrm{H}},u_3^{\mathrm{H}})$	−0.9981
$cc(u_1^{\mathrm{Li}},u_2^{\mathrm{Li}})$	−0.9915		

other terms of the expansion for electric dipole moment (2)

$$\frac{\mathrm{d}\mu(x)}{\mathrm{d}x} = \left[eR_0(1+x)/2\right] \sum_{i=0} \left(s_i^{\mathrm{a}} - s_i^{\mathrm{b}}\right) x^i, \quad (28)$$

at assumed polarity $^{+}\mathrm{AB}^{-}$.

By equating sets of coefficients of x in (10) and (28) to a common exponent one generates relationships

$$s_{k-1}^{\mathrm{a}} - s_{k-1}^{\mathrm{b}} = \left(t_{k-1}^{\mathrm{a}} - t_{k-1}^{\mathrm{b}} + t_k^{\mathrm{a}} - t_k^{\mathrm{b}}\right)k, \quad k = 1,2,3\ldots, \quad (29)$$

valid for Dunham's expansion of pertinent radial functions.

Having derived (20) one can test both its validity and physical significance of evaluated nonadiabatic vibrational and rotational parameters of LiH. Although the latter were assigned for radial functions $\alpha(R)$ and $\beta(R)$ expanded in Ogilvie's variable, the first relation from (29)

$$s_0^{\mathrm{Li}} - s_0^{\mathrm{H}} = t_0^{\mathrm{Li}} - t_0^{\mathrm{H}} + t_1^{\mathrm{Li}} - t_1^{\mathrm{H}}, \quad (30)$$

is valid within a linear approximation

$$z = \frac{2(R-R_0)}{R+R_0} = \frac{x}{1+x/2} \approx x(1-x/2) \quad (31)$$

also for Ogilvie's expansion. Hence one can calculate from (30) $s_0^{\mathrm{H}} = 0.540(68)$ which conforms acceptably with $s_0^{\mathrm{H}} = 0.618(66)$ from the fit. The error of calculated s_0^{H} propagates from errors of fitted parameters t_0^{Li}, t_0^{H} and t_1^{H}.

Only one coefficient s_0^{H} pertaining to the radial function $\beta(R)$ can be assigned within the proposed approach at present; for the present data parameters s_0^{Li} and $s_{i>0}^{\mathrm{Li,H}}$ are redundant. The test calculation by means of (30) shows that $s_0^{\mathrm{Li}} \approx 0$; we conclude that nonadiabtic vibrational effects for the relatively heavier lithium center are negligible. The same holds for nonadiabatic rotational effects of higher order represented by $t_{i>0}^{\mathrm{Li}}$.

These results demonstrate that from highly resolved infrared and microwave spectra one can derive useful information about electric and magnetic properties of free diatomic molecules. The former are evaluated from only spectra of samples measured in the absence of externally applied electric or magnetic field. The results obtained for LiH and heavier molecules [8,12,13] indicate that not only magnitudes but also signs of pertinent quantities with respect to molecular frames may be deduced from the spectra; this information is difficult to acquire otherwise. We have a new source of meaningful information on electric and magnetic properties of diatomic systems, derivable by inversion of spectral data.

Application of this analytic version of DS-cP to evaluate from spectra nonadiabatic vibrational effects of higher order, and details concerning calculation of R_J with Maple will be published elsewhere.

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